

PROJECT SYNOPSIS

On

An AI-Based Application for Entrance Exam Preparation and Habit Building

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Computer Application

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1 Introduction

Entrance exams for postgraduate computer application programs, such as NIMCET and PG CET-MCA, require students to prepare across multiple subjects within a limited timeframe. In practice, this preparation is often fragmented — students rely on a mix of scattered YouTube videos, unrelated PDF question banks, and generic doubt-solving forums, with no single tool tying their progress together or adapting to what they specifically need to work on. At the same time, most students preparing for competitive exams also struggle with the habits and consistency needed to sustain months of preparation, a concern that is rarely addressed by any of the tools built for exam content itself.

This project addresses both sides of that gap. It is an AI-assisted mobile and web application that combines focused entrance exam preparation with a lightweight personal development layer, giving students one place to both study effectively and build the daily consistency that exam preparation demands.

2 About the Project

The application is built around two modes, switchable within the same account, so a student can move between personal development and focused exam preparation without leaving the app.

2.1 Default Mode

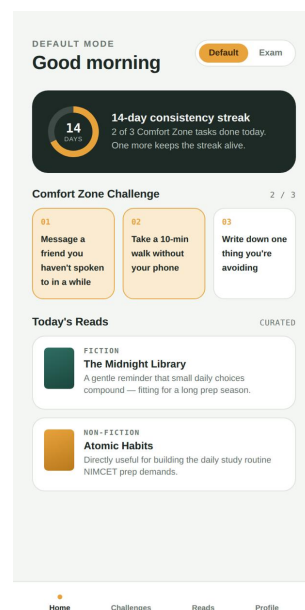


Figure 1: User interface of the Default Mode focusing on personal development.

Default Mode is the application's personal development layer, designed to build consistency and habits alongside academic preparation. It includes a daily "Comfort Zone Challenge" of three short, curated tasks that gently encourage the user outside routine behaviour, along with daily book recommendations — one fiction and one non-fiction title — each paired with a short, concrete reason for the suggestion.

2.2 Exam Mode

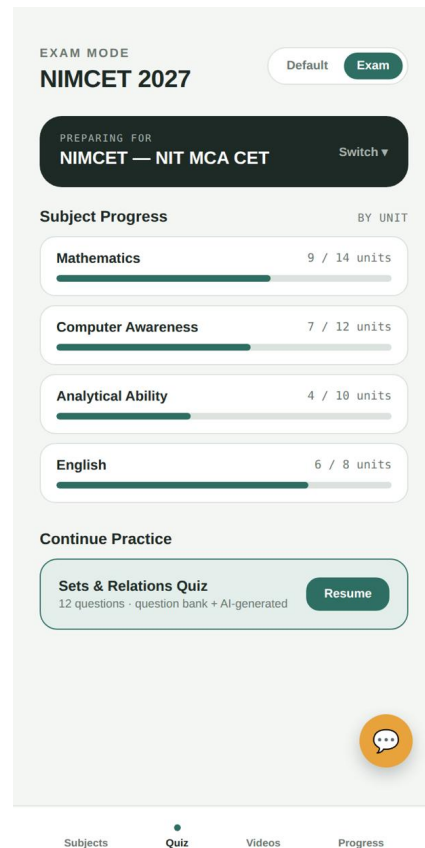


Figure 2: User interface of the Exam Mode customized for academic preparation.

Exam Mode is the application’s academic core. A student selects the specific entrance exam they are preparing for, which loads a corresponding set of subjects and syllabus units. Within this mode, the application provides a quiz system that begins with a predefined question bank and later incorporates live AI-generated questions, an AI chatbot for doubt-solving and concept explanations, curated YouTube video recommendations mapped to syllabus units, and unit-by-unit progress tracking across each subject.

3 Purpose of the Project

Most existing exam-preparation tools focus narrowly on test content — question banks, mock tests, or video lectures — without any attention to the habits and mindset that actually determine whether a student studies consistently. At the same time, most personal-development or habit-tracking apps are entirely generic and disconnected from a student’s actual academic goals. The purpose of this project is to bring both of these needs into a single, coherent application: a focused, AI-assisted preparation tool for a specific set of entrance exams, paired with a lightweight, non-intrusive habit and personal-growth layer.

4 Objectives

- To build a single-codebase application that runs on both web and mobile using React Native and Expo. ****future enhancement****
- To allow students to select a specific entrance exam and receive subject and unit content tailored to that exam's syllabus.
- To provide low-cost, low-latency quiz practice using a predefined question bank before introducing live AI-generated questions.
- To provide an AI chatbot capable of answering syllabus-specific doubts and explaining concepts on demand.
- To track a student's progress at the level of individual syllabus units, not just overall subjects.
- To encourage habit-building and personal growth through small, human-curated daily tasks and reading recommendations.

5 Hardware and Software Requirements

5.1 Hardware Requirements

- **Development machine:** Any laptop/desktop capable of running Node.js and a code editor.
- **Testing device:** A smartphone (Android/iOS) with the Expo Go app, or an emulator/simulator.
- **Internet connection:** Required for cloud database access, AI API calls, and package installation.

5.2 Software Requirements

- **Operating System:** Windows / macOS / Linux (any OS supporting Node.js).
- **Code Editor:** Cursor.
- **Runtime:** Node.js (LTS version).
- **Framework:** React+Vite (managed workflow).
- **Backend & Database:** Supabase (PostgreSQL, Authentication, Edge Functions).
- **Version Control:** Git and GitHub.
- **Build & Deployment:** Expo Application Services (EAS)/Cloudflare workers

6 Technologies Used

6.1 Frontend

Technology	Description
Figma	Design and prototype tool used to design user interface and user experience.
React	Used to create a dynamic web application.
React Router	Standard routing library used for seamless navigation between different screens and components within the application.
React Context	Built-in state management tool used to efficiently share global data (such as user authentication or current exam mode) across the application.
Vite	A fast, modern build tool and development server used to bundle the frontend application with optimized performance.
React Native	Cross-platform UI framework for mobile and web.
Expo	Managed workflow, build tooling, and OTA updates for React Native.

Note: React Native and Expo are used to create cross-platform applications and are currently categorized into future enhancements.

6.2 Backend and Data

Technology	Description
Supabase	Managed PostgreSQL database, authentication (Email/Password and Google Sign-In), and serverless Edge Functions.
Zod	A TypeScript-first schema declaration and validation library used to ensure data integrity and runtime type safety across the backend.

6.3 Artificial Intelligence

Technology	Description
Claude API	Used for the chatbot and concept explanations, which require nuanced, high-quality reasoning.
Gemini API	Used for high-volume, templated tasks such as quiz generation and book-recommendation reasoning.

Note: For the Gemini API, we are utilizing a Bring Your Own Key (BYOK) approach. This allows the application to use user-provided API keys, which ensures secure, decentralized access and efficiently manages API costs and rate limits during high-volume quiz generation.

6.4 Supporting Services

Technology	Description
YouTube Data API	For topic-relevant video recommendations.
YouTube Iframe Player Api	For embedded in-app video playback and watch progress tracking
Cloudflare workers	web hosting and deployment

6.5 Development/Testing Tools

Technology	Description
Cursor	AI-assisted code editor used for the majority of implementation work.
Git and GitHub	Version control and source management.
ESLint + Prettier	Used for maintaining code quality, enforcing consistent formatting, and catching syntax errors during development.
Vitest	A fast, lightweight testing framework used to write and run unit tests to ensure application logic and reliability.
Wrangler	The official command-line interface (CLI) tool used for developing, testing locally, and deploying the application to Cloudflare.

6.6 Programming Languages

Technology	Description
JavaScript	The primary programming language used for building both the interactive frontend and backend logic of the application.
TypeScript	A strongly typed superset of JavaScript used to enhance code reliability, catch errors early, and improve maintainability during development.
CSS / Tailwind CSS	Used for designing and styling the application, ensuring a modern, clean, and responsive user interface across all devices.
SQL	Structured Query Language used for defining the schema, querying, and managing relational data within the PostgreSQL database.

7 Conclusion

This project demonstrates that a focused, well-scoped application can meaningfully support entrance exam preparation without the overhead of a full production ed-tech platform. By pairing subject-wise quiz practice and AI-assisted doubt-solving with a lightweight habit-building layer, the application addresses both the academic and consistency challenges students face during exam preparation — giving it a distinct identity beyond a standard quiz tool.

8 Future Enhancements

- **Retrieval-Augmented Generation (RAG):** Grounding chatbot answers in specific textbook or course material using a vector database, once real usage shows generic AI responses are insufficient.
- Spaced-repetition style reminders that nudge students to revisit syllabus units they previously scored poorly on.
- Detailed, per-question explanations for quiz answers, not just a right/wrong indicator.
- Support for tracking preparation across multiple exams simultaneously, for students targeting more than one entrance test.
- converting it to a mobile application.
- A lightweight, privacy-conscious social/accountability layer showing streaks (not content) between friends.
- Expansion of the predefined question bank patterns.

9 System Architecture

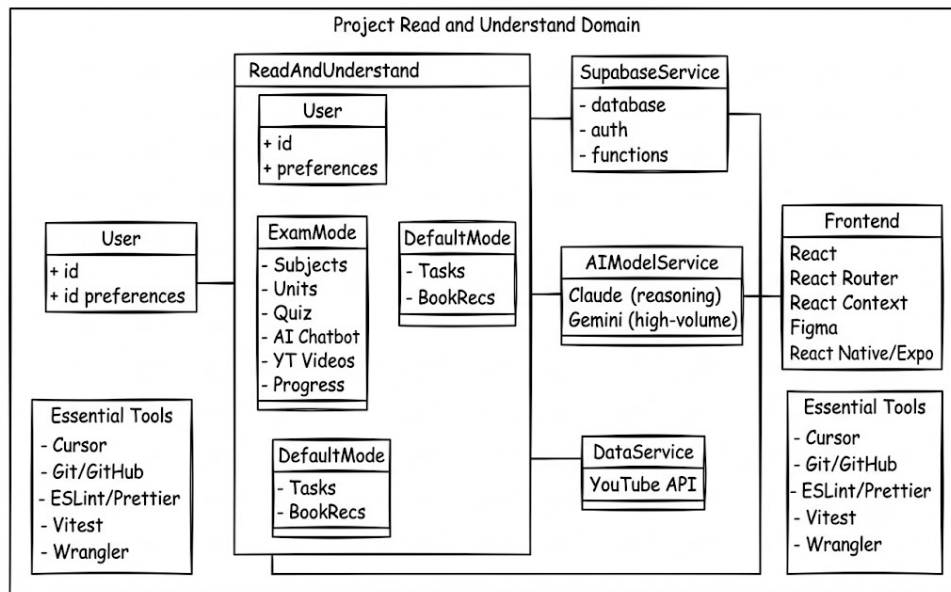


Figure 3: system architecture of the application.

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